

# Long-Horizon Momentum-Based Multi-Strategy Trading System

Synthetic data construction, validation diagnostics, and historical backtest synthesis

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## Abstract

This document analyzes a tactical multi-strategy investment system built from four allocation blocks: Momentum Stocks, Gold, Sector, and Bond. The detailed trading rules are not reproduced here; they are delegated to the companion strategy specification. This report focuses instead on the construction of the long synthetic histories, the validation of the sector and bond data extensions, the individual backtests on their maximum available samples, and the combined portfolio backtest from January 1980 onward, which is the first common start date across the complete strategy set.

## 1 Introduction

The strategy is designed as a long-horizon tactical allocation system composed of four return streams: Momentum Stocks, Gold, Sector, and Bond. The signal definitions, ranking rules, position-selection logic, and implementation details of each block are documented in the companion PDF [1]. This document deliberately avoids restating those rules and concentrates on the data-engineering work needed to test the blocks over long historical samples.

The portfolio is organized around four allocation blocks rather than eight separate strategy components. The five regional stock momentum portfolios are consolidated into a single *Momentum Stocks* block. This makes the allocation more readable while preserving the internal geographical diversification of the original implementation.

The strategy aims, approximately and over long investment horizons, to achieve an annual gain around 20%, a Sharpe ratio close to 2, and a maximum drawdown around 10%. These are strategic design objectives rather than guaranteed outcomes. Realized performance depends on market regimes, data quality, execution costs, taxes, slippage, liquidity, and the exact universe used.

## 2 Portfolio Overview

The final strategy uses four monthly return streams. The stock momentum block receives half of the portfolio weight and internally aggregates the five regional stock momentum portfolios. Gold, Sector, and Bond blocks provide diversifying exposure to different macro and asset-class regimes.

**Table 1:** Strategy allocation blocks and target portfolio weights.

| Allocation block | Role in the combined portfolio                                                                                                           | Portfolio weight |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Momentum Stocks  | Consolidated return stream generated from five regional stock universes. The selection rules are defined in the companion specification. | 50%              |
| Gold             | Tactical gold/equity return stream generated by the companion specification and tested on a long reconstructed data set.                 | 10%              |
| Sector           | U.S. sector-timing return stream generated by the companion specification and tested on reconstructed sector histories.                  | 10%              |
| Bond             | Fixed-income tactical return stream generated by the companion specification and tested on reconstructed bond histories.                 | 30%              |

### 3 Data Construction and Preprocessing

The backtest operates at monthly frequency. Daily or monthly source data are converted to calendar month-end observations, and all return series are compounded from month to month. Whenever possible, total-return or adjusted-close data are used so that dividends, distributions, splits, and other corporate actions are reflected in the historical return stream.

#### 3.1 Momentum Stocks Data

The Momentum Stocks block starts in January 1980. The five regional investment universes are defined from locally saved TradingView component pages, but the historical price source for the stock backtest is EODHD. TradingView is therefore used to define the investment lists, while EODHD provides the adjusted daily historical prices used by the strategy [2].

Operationally, each TradingView component page is opened in Google Chrome, the full list of constituents is loaded with the page’s *Load more* button, and the complete page is saved locally in MHTML format. The parser extracts the listed securities, filters them by listing currency, and maps exchange codes into data-provider tickers. EODHD adjusted daily closes are then retrieved for the resulting ticker lists. The five regional portfolios are finally aligned at month end and aggregated into the consolidated Momentum Stocks block.

#### 3.2 Gold and S&P 500 Total Return Data

The Gold block requires two long monthly total-return histories: a gold leg and an equity leg. The modern tradable implementation is expressed with GLD and SPY, but these ETFs do not cover the full research sample. A synthetic long-history data set is therefore built so that the Gold block can be tested before the ETF era.

For the gold leg, the historical gold price is collected as a month-end price series. The preferred observation is the last available trading day or quotation of each calendar month, not a monthly average. The series is first cleaned for missing values and then rebased into a continuous wealth index. When GLD adjusted-close data become available, the synthetic gold series is linked to GLD over the overlapping period by matching the level at the splice date. Since physical gold does not pay dividends, the main adjustment is the continuity of the price level and the consistency of the monthly calendar.

For the equity leg, the objective is to represent the S&P 500 as a total-return investment rather than as a price-only index. The long history is reconstructed from the S&P 500 price index and dividend information, using the Shiller long-term stock-market database as the dividend reference

[4]. Monthly total returns are obtained by combining price appreciation with reinvested dividends. The resulting synthetic total-return index is then linked to SPY adjusted-close data when the ETF becomes available. The adjusted-close ETF history is used because it incorporates distributions and splits.

The two reconstructed legs are finally aligned on the same month-end calendar and rebased to a common starting value. The relative level of the gold total-return proxy versus the S&P 500 total-return proxy is retained as an input to the externally specified Gold block. This guarantees that the pre-ETF backtest and the modern ETF implementation use the same economic objects: gold exposure on one side and broad U.S. equity total return on the other.

### 3.3 Sector Data Reconstruction Since 1980

The Sector block requires monthly histories for the sector ETFs used in the inflation proxy and in the regime allocation map. The tradable ETF implementation is based on SPY and the Select Sector ETFs XLB, XLE, XLF, XLI, XLK, XLP, XLU, and XLV. For validation and for a complete U.S. sector data set, the reconstruction also includes Consumer Discretionary and Communication Services, mapped respectively to XLY and XLC. The difficulty is that the sector ETFs mostly begin in 1998–1999, while the research sample used by the multi-strategy portfolio begins in January 1980. The reconstruction therefore uses long Fama–French U.S. industry returns as the historical backbone and modern sector ETFs as observable anchors over the common period.

The objective is to create economically consistent monthly total-return proxies, not to claim that the pre-inception series are official GICS sector indices. The reconstructed histories are research proxies: they are intended to preserve broad sector risk premia, cyclical relationships, and relative performance dynamics, while remaining compatible with the ETF implementation that can be traded today.

#### 3.3.1 Long-history source: Fama–French 10 industry portfolios

The long source is the Fama–French 10 industry portfolio data set [3]. It provides monthly U.S. industry returns over a much longer sample than the sector ETFs. The 10 industry blocks used by the reconstruction are

$$\mathcal{K} = \{\text{NoDur}, \text{Durbl}, \text{Manuf}, \text{Enrgy}, \text{HiTec}, \text{Telcm}, \text{Shops}, \text{Hlth}, \text{Utils}, \text{Other}\}. \quad (1)$$

The source file gives monthly returns in percentage units and with dates encoded as YYYYMM. Each line is converted into a calendar month-end date, and returns are transformed into decimal form. For each Fama–French industry  $k$ , a base-100 wealth index is reconstructed by compounding:

$$I_t^{FF,k} = 100 \prod_{u \leq t} \left( 1 + r_u^{FF,k} \right), \quad (2)$$

where  $r_u^{FF,k}$  is the decimal monthly return of industry  $k$ . Since the Fama–French research returns include dividends according to the CRSP convention used in the data library, the resulting series are total-return-like industry wealth indices.

#### 3.3.2 Economic mapping from Fama–French industries to sectors

The Fama–French industries do not exactly match the modern GICS sectors represented by the Select Sector ETFs. A static economic mapping is therefore applied. The mapping favors robust economic interpretation over a point-in-time replication of a changing classification system. This is especially important for Communication Services, which experienced a major GICS reclassification

in 2018, and for Health Care and Consumer Staples, which do not map perfectly into broad industry blocks.

Let  $w_{s,k}$  be the weight of Fama–French industry  $k$  in synthetic sector  $s$ . The sector return before splicing is computed as a linear combination of monthly industry returns:

$$r_t^{\text{syn},s} = \sum_{k \in \mathcal{K}} w_{s,k} r_t^{\text{FF},k}, \quad \sum_{k \in \mathcal{K}} w_{s,k} = 1. \quad (3)$$

The synthetic sector level is then recomposed by compounding:

$$I_t^{\text{syn},s} = 100 \prod_{u \leq t} (1 + r_u^{\text{syn},s}). \quad (4)$$

**Table 2:** Static mapping from Fama–French 10 industries to synthetic U.S. sectors.

| Synthetic sector       | Fama–French combination                                                                  |
|------------------------|------------------------------------------------------------------------------------------|
| Materials              | Manuf: 0.70, Enrgy: 0.10, Other: 0.20                                                    |
| Energy                 | Enrgy: 1.00                                                                              |
| Financials             | Shops: 0.10, Other: 0.90                                                                 |
| Industrials            | Manuf: 0.75, Other: 0.25                                                                 |
| Technology             | HiTec: 0.90, Telcm: 0.10                                                                 |
| Consumer Staples       | NoDur: 0.85, Shops: 0.15                                                                 |
| Utilities              | Utils: 1.00                                                                              |
| Health Care            | Durbl: 0.01, Manuf: 0.00, HiTec: 0.06, Telcm: 0.01, Shops: 0.00, Hlth: 0.87, Other: 0.05 |
| Consumer Discretionary | NoDur: 0.10, Durbl: 0.35, Shops: 0.45, Other: 0.10                                       |
| Communication Services | Durbl: 0.02, HiTec: 0.26, Telcm: 0.63, Shops: 0.10                                       |

The mapping intentionally separates three different objects: the long industry returns, the modern sector ETFs, and the final synthetic histories. This avoids confusing a research proxy with an official retro-calculated sector index. The `synthetic_only` files contain the pure Fama–French-based histories. The `spliced` files use ETF adjusted-close histories when available and extend the earlier part with the synthetic proxy.

### 3.3.3 Monthly synchronization and return convention

All series are placed on a single month-end grid. Fama–French monthly observations are converted from YYYYMM to the last calendar day of the month. ETF price histories from Yahoo are sampled at the last available traded close of each month and then assigned to the same calendar month-end date. The portfolio weights generated at the end of month  $t$  are applied only to the return of month  $t + 1$  by shifting them forward by one month. This timing convention is essential because the strategy must not use a return that was not yet known when the signal was formed.

The monthly ETF return for sector  $s$  is computed from adjusted-close levels:

$$r_t^{\text{ETF},s} = \frac{A_t^{\text{ETF},s}}{A_{t-1}^{\text{ETF},s}} - 1, \quad (5)$$

where  $A_t^{\text{ETF},s}$  is the month-end adjusted-close level. Adjusted-close histories are used because they incorporate splits and cash distributions, making them comparable to total-return-like Fama–French returns.

### 3.3.4 Splicing with observed sector ETFs

For each sector  $s$ , the synthetic proxy is compared with its corresponding ETF over the common sample. The ETF is not used to rewrite the pre-inception history month by month. Instead, it provides an observable modern anchor and a validation target. The reconstruction keeps the synthetic historical returns intact and, when the `spliced` version is requested, links the old synthetic level to the modern ETF level at a joining date.

Let  $\tau_s$  be the first month where a usable ETF adjusted-close observation exists. The spliced sector level is

$$I_t^{\text{splice},s} = \begin{cases} I_t^{\text{syn},s} \frac{A_{\tau_s}^{ETF,s}}{I_{\tau_s}^{\text{syn},s}}, & t < \tau_s, \\ A_t^{ETF,s}, & t \geq \tau_s. \end{cases} \quad (6)$$

Only a constant scaling factor is applied before  $\tau_s$ . The old synthetic returns are therefore not distorted by the level match. This is preferable to a pure level adjustment or a regression fit because the backtest uses monthly returns and rolling relative levels; preserving the monthly return dynamics is more important than forcing a perfect visual match of index levels.

### 3.3.5 S&P 500 total-return reconstruction used by the sector block

The Sector block also requires a long equity market proxy for the growth regime test. The benchmark is not treated as a price-only S&P 500 series. It is reconstructed as an approximate total-return index from the monthly level of  $\hat{\text{GSPC}}$  and the historical dividend information from the Shiller data set [4]. If  $P_t$  is the S&P 500 price level, the price return is

$$r_t^{\text{price}} = \frac{P_t}{P_{t-1}} - 1. \quad (7)$$

If  $D_t$  is the annual dividend amount from Shiller, the annual dividend yield is approximated by

$$y_t^{\text{div}} = \frac{D_t}{P_t}, \quad (8)$$

and the monthly dividend return is approximated as

$$r_t^{\text{div}} \approx \frac{y_t^{\text{div}}}{12}. \quad (9)$$

The reconstructed monthly total return is then

$$r_t^{SP500TR} = \left(1 + r_t^{\text{price}}\right) \left(1 + r_t^{\text{div}}\right) - 1. \quad (10)$$

Finally, the S&P 500 total-return proxy is compounded as a base-100 wealth index:

$$I_t^{SP500TR} = 100 \prod_{u \leq t} \left(1 + r_u^{SP500TR}\right). \quad (11)$$

This benchmark is sufficiently consistent for long monthly research, but it remains an approximation of the official S&P 500 Total Return index. The official index uses exact dividend ex-dates, exact index composition, corporate actions, and licensed S&P methodology.

### 3.3.6 Validation against sector ETFs

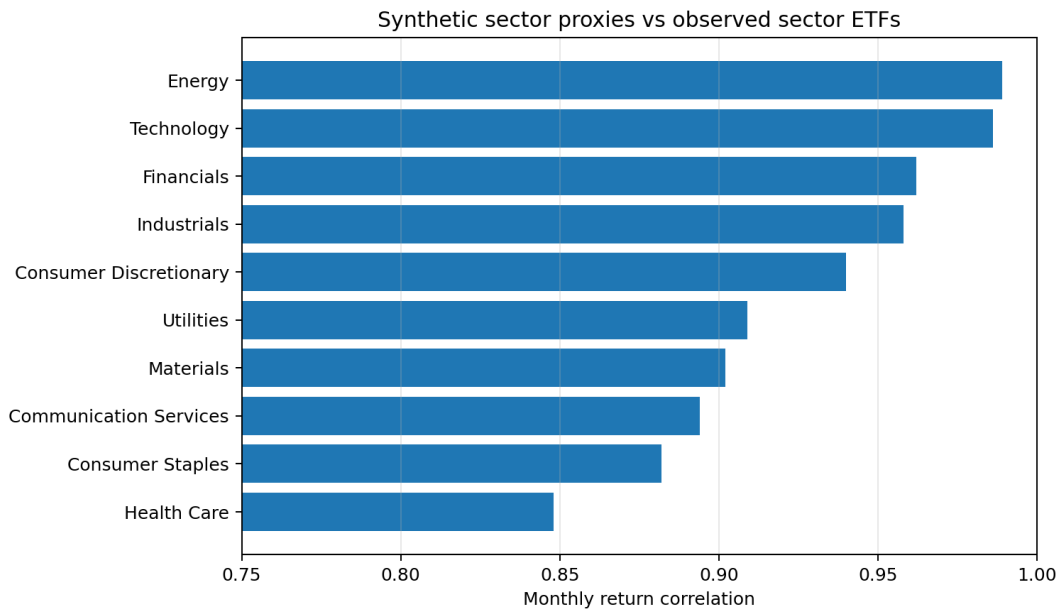
The quality of each synthetic proxy is measured on the ETF overlap period. The primary diagnostic is the monthly return correlation,

$$\rho_s = \text{corr} \left( r_t^{\text{syn},s}, r_t^{\text{ETF},s} \right), \quad (12)$$

and the secondary diagnostic is the annualized tracking error,

$$TE_s = \sqrt{12} \text{ stdev} \left( r_t^{\text{syn},s} - r_t^{\text{ETF},s} \right). \quad (13)$$

The overlap starts in January 1999 for most Select Sector ETFs and in July 2018 for XLC. The shorter XLC history makes Communication Services structurally less reliable before 2018 than the older sectors.



**Figure 1:** Monthly return correlations between synthetic sector proxies and observed sector ETFs on the common sample.

**Table 3:** ETF validation statistics for the synthetic sector proxies.

| Sector                 | ETF | Months | Start      | End        | Corr. | Ann. TE |
|------------------------|-----|--------|------------|------------|-------|---------|
| Materials              | XLB | 327    | 1999-01-31 | 2026-03-31 | 0.902 | 9.13%   |
| Energy                 | XLE | 327    | 1999-01-31 | 2026-03-31 | 0.989 | 3.72%   |
| Financials             | XLF | 327    | 1999-01-31 | 2026-03-31 | 0.962 | 6.23%   |
| Industrials            | XLI | 327    | 1999-01-31 | 2026-03-31 | 0.958 | 5.39%   |
| Technology             | XLK | 327    | 1999-01-31 | 2026-03-31 | 0.986 | 3.81%   |
| Consumer Staples       | XLP | 327    | 1999-01-31 | 2026-03-31 | 0.882 | 6.06%   |
| Utilities              | XLU | 327    | 1999-01-31 | 2026-03-31 | 0.909 | 6.43%   |
| Health Care            | XLV | 327    | 1999-01-31 | 2026-03-31 | 0.848 | 7.82%   |
| Consumer Discretionary | XLY | 327    | 1999-01-31 | 2026-03-31 | 0.940 | 6.66%   |
| Communication Services | XLC | 93     | 2018-07-31 | 2026-03-31 | 0.894 | 8.59%   |

Energy, Technology, Financials, and Industrials show very high overlap correlations, which is consistent with their relatively clean mapping to Fama–French industries. Consumer Staples and Health Care are less exact because broad research-industry groups do not reproduce modern ETF

composition perfectly. Communication Services must be interpreted with extra caution because the modern sector and the XLC ETF have a short live history and a classification break around 2018.

### 3.3.7 Robustness check on the ETF overlap

The ETF overlap is also split internally into a training segment and a validation segment. This check is kept as a robustness control rather than as a second table, because it uses the same two diagnostics as Table 3: monthly correlation and annualized tracking error. The split confirms that the main sector proxies remain stable out of sample. Validation correlations remain close to, or above, their training values for the major proxies, especially Energy, Financials, Industrials, Technology, Consumer Staples, Utilities, Health Care, and Consumer Discretionary. Communication Services remains the weakest case because XLC has only a short live history and the GICS definition of the sector changed materially around 2018.

### 3.3.8 Operational generation workflow

The historical sector data set is generated with the following operational sequence:

1. download or load the Fama–French 10 industry monthly returns;
2. convert returns from percentages to decimals and convert YYYYMM rows to calendar month-end dates;
3. build base-100 industry wealth indices by compounded returns;
4. apply the static mapping in Table 2 to obtain monthly synthetic sector returns;
5. compound the synthetic returns into `synthetic_only` sector indices;
6. download sector ETF adjusted-close histories, sample them at month end, and compute ETF monthly returns;
7. compare each proxy with its ETF on the common sample and export correlation, tracking-error, and train/validation diagnostics;
8. build `spliced` histories by using ETF adjusted-close data in the live ETF period and level-linked Fama–French proxies before ETF inception;
9. reconstruct the S&P 500 total-return proxy from `~GSPC` and Shiller dividends and align it to the same month-end grid;
10. export synchronized sector levels, sector returns, S&P 500 total-return components, validation diagnostics, and SectorStrategy backtest inputs.

The workflow produces the pure synthetic histories, the spliced histories, synchronized monthly returns, S&P 500 total-return components, ETF-validation diagnostics, and the sector backtest input/output files. These files are intermediate audit artifacts rather than additional methodological steps, so they are not reproduced as a separate table.

### 3.3.9 Interpretation limits

The pre-ETF sector histories should be used as economic proxies rather than as observations of tradable sector ETFs. The period before 1999 cannot be validated directly against the Select

Sector ETF universe. It tests whether sector-like risk premia would have remained robust under long historical U.S. industry returns. The post-1999 validation supports the mapping, but it cannot eliminate the uncertainty attached to earlier decades.

The reconstructed S&P 500 total-return series is also an approximation. It is appropriate for long monthly backtests, but it should not be presented as a licensed official S&P DJI total-return index. For institutional publication, the same methodology should be checked against an official S&P Total Return, CRSP, or Bloomberg source over the longest overlapping sample.

### 3.4 Bond Data Reconstruction Since 1971

The Bond block is reconstructed from January 1971 onward. The tradable universe is composed of SHY, IEF, LQD, HYG, TIP, and CWB, representing short-duration Treasuries, intermediate Treasuries, investment-grade corporate credit, high-yield credit, inflation-linked bonds, and convertible bonds. Since these ETFs were launched at different dates and most of them do not exist during the early part of the sample, each sleeve is extended backward with a synthetic monthly total-return proxy. The objective is not to reproduce proprietary benchmark indices exactly, but to create long, homogeneous, economically interpretable histories suitable for monthly backtesting.

All raw interest-rate series are sampled at month end and converted from annual percentages into decimals before being used in the formulas. The synthetic indices are arbitrary wealth indices rebased to one; therefore, only their monthly returns are used by the strategy. The non-investable T10Y3M variable is kept separate from the six investable sleeves and is used only as a regime filter.

#### 3.4.1 Core return notation

Let  $P_{i,t}$  denote the synthetic month-end wealth index of asset  $i$ . The simple monthly return is

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1, \quad (14)$$

and a wealth index is rebuilt from returns by compounding:

$$P_{i,t} = P_{i,0} \prod_{u=1}^t (1 + r_{i,u}), \quad (15)$$

with  $P_{i,0} = 1$  in the reference data files.

#### 3.4.2 Rolled par-bond approximation

Treasury sleeves and the bond components of the credit sleeves are reconstructed with a rolled par-bond approximation. At the start of each month, the bond is assumed to trade at par, with an annual coupon equal to the yield observed at the previous month-end. At the end of the month, the same bond is revalued at the new market yield, with one month less to maturity. This is more explicit than a linear duration approximation and gives a direct price-and-carry decomposition.

For a bond with remaining maturity  $T$ , annual coupon  $c$ , yield to maturity  $y$ , face value 100, and semi-annual coupons, the price approximation is

$$B(c, y, T) = \frac{100 c/2}{y/2} \left( 1 - \left( 1 + \frac{y}{2} \right)^{-2T} \right) + 100 \left( 1 + \frac{y}{2} \right)^{-2T}. \quad (16)$$

The monthly total return of the rolled bond is then

$$r_t^{\text{bond}} = \frac{B(y_{t-1}, y_t, T - \frac{1}{12}) + 100 \frac{y_t - 1}{12}}{100} - 1. \quad (17)$$



The first term revalues the bond after the yield move, while the second term approximates one month of coupon carry.

### 3.4.3 SHY and IEF Treasury sleeves

SHY represents short-maturity U.S. Treasury exposure. It is reconstructed from the 3-year Treasury constant maturity yield  $GS3_t$  with an effective maturity of two years:

$$r_t^{\text{SHY},0} = r^{\text{bond}}(GS3_t, T = 2)_t. \quad (18)$$

This proxy is intended to behave like a 1–3 year Treasury allocation and also serves as the safe-haven asset when the yield-curve filter is negative.

IEF represents intermediate Treasury exposure, approximately the 7–10 year segment. It is reconstructed from the 10-year Treasury constant maturity yield  $GS10_t$ , using an effective maturity of 10.5 years. A small annual basis correction of 40 basis points is added, i.e.  $0.004/12$  per month:

$$r_t^{\text{IEF},0} = r^{\text{bond}}(GS10_t, T = 10.5)_t + \frac{0.004}{12}. \quad (19)$$

The correction absorbs persistent differences between a single constant-maturity synthetic bond and the actual ETF benchmark, which is a diversified rolled basket of Treasury bonds.

### 3.4.4 LQD investment-grade credit sleeve

LQD represents U.S. investment-grade corporate credit. Because long histories of the ETF benchmark itself are not public before ETF inception, a synthetic investment-grade yield is built from Moody's Aaa and Baa corporate yields:

$$y_t^{IG} = 0.80 \text{ Aaa}_t + 0.20 \text{ Baa}_t. \quad (20)$$

The base bond component is a 10-year rolled par bond:

$$r_t^{IG,\text{bond}} = r^{\text{bond}}(y_t^{IG}, T = 10)_t. \quad (21)$$

The investment-grade spread proxy is

$$s_t^{IG} = y_t^{IG} - GS10_t. \quad (22)$$

A spread-duration term is added to make the proxy respond to credit-spread widening and tightening. A small equity-beta component captures liquidity and cyclical credit effects, and a conservative carry haircut is subtracted:

$$r_t^{\text{LQD},0} = r_t^{IG,\text{bond}} - 10 \Delta s_t^{IG} + 0.08 r_t^{SP500} - \frac{0.015}{12}. \quad (23)$$

In the common universe starting in 1971, the S&P 500 return series is available and is therefore included in the credit proxy.

### 3.4.5 HYG high-yield credit sleeve

HYG represents U.S. high-yield corporate credit. Since freely available high-yield total-return histories do not extend sufficiently far back, Moody's Baa yield is used as a long-history credit anchor. The base bond component is a shorter four-year rolled par bond:

$$r_t^{\text{Baa},\text{bond}} = r^{\text{bond}}(\text{Baa}_t, T = 4)_t. \quad (24)$$

The high-yield spread proxy is

$$s_t^{HY} = BAA_t - GS10_t. \quad (25)$$

The raw high-yield proxy return is then

$$r_t^{HYG,0} = r_t^{BAA,bond} - 8 \Delta s_t^{HY} + 0.30 r_t^{SP500} - \frac{0.03}{12}. \quad (26)$$

The shorter maturity, spread sensitivity, equity beta, and higher haircut are designed to reproduce the hybrid behavior of high-yield bonds: coupon carry, credit-spread risk, and exposure to the equity cycle.

### 3.4.6 TIP inflation-linked bond sleeve

TIP is handled separately because U.S. TIPS did not exist before 1997 and directly observable real-yield data begin only later. The retained series is a nominal total-return proxy for inflation-linked bonds, not a real return net of inflation.

For 1971–1973, the TIP proxy is set equal to the intermediate Treasury proxy:

$$r_t^{TIP,0} = r_t^{IEF,0}, \quad 1971 \leq t \leq 1973. \quad (27)$$

This convention follows the idea that before a dedicated inflation-linked market is available, intermediate Treasuries are the closest defensive fixed-income substitute.

For 1974–2002, the proxy combines realized CPI inflation with a synthetic real yield. Monthly realized inflation is

$$\pi_t = \frac{CPI_t}{CPI_{t-1}} - 1. \quad (28)$$

Expected inflation is approximated by trailing 12-month inflation, denoted  $\hat{\pi}_t^{12m}$ , and the synthetic real yield is

$$y_t^R = GS10_t - \hat{\pi}_t^{12m}. \quad (29)$$

The nominal total return of the TIPS proxy is approximated by

$$r_t^{TIP,0} = \pi_t + \frac{y_{t-1}^R}{12} - D_{TIPS} \Delta y_t^R + \frac{0.006}{12}, \quad D_{TIPS} = 7. \quad (30)$$

The four terms correspond respectively to principal indexation, real coupon carry, price sensitivity to changes in real yields, and a 60-basis-point annual basis correction.

From 2003 onward, the observed 10-year TIPS real yield  $DFII10_t$  is used. Monthly inflation compensation is approximated by the 10-year breakeven rate divided by 12:

$$\pi_t^{BE} = \frac{GS10_t - DFII10_t}{12}. \quad (31)$$

The monthly return becomes

$$r_t^{TIP,0} = \pi_t^{BE} + \frac{DFII10_{t-1}}{12} - D_{TIPS} \Delta DFII10_t + \frac{0.006}{12}. \quad (32)$$

The complete TIP proxy is obtained by concatenating the three subperiods.

### 3.4.7 CWB convertible-bond sleeve

CWB represents U.S. convertible bonds. Convertibles have a hybrid profile that is partly equity-like and partly credit-like. The long-history proxy is therefore built as a convex combination of S&P 500 returns and the HYG proxy returns:

$$r_t^{\text{CWB},0}(w) = w r_t^{\text{SP500}} + (1 - w) r_t^{\text{HYG}}. \quad (33)$$

Several equity weights were tested,

$$w \in \{0.60, 0.65, 0.70, 0.75, 0.80\}, \quad (34)$$

and the retained reference series is

$$r_t^{\text{CWB},0} = 0.60 r_t^{\text{SP500}} + 0.40 r_t^{\text{HYG}}. \quad (35)$$

This weight provided the best compromise between correlation and tracking error over the common sample with the CWB ETF, while preserving a conservative credit contribution.

### 3.4.8 Calibration against observed ETFs

The raw proxies are economically meaningful but imperfect. They are therefore calibrated over the period where the corresponding ETFs are observable. The calibration is parsimonious: it does not try to match every monthly ETF return, but rather the main long-run statistics used in the backtest.

For a raw proxy return  $r_t^0$ , the calibrated return is

$$r_t = \alpha + \beta_+ r_t^0 \quad \text{if } r_t^0 \geq 0, \quad r_t = \alpha + \beta_- r_t^0 \quad \text{if } r_t^0 < 0. \quad (36)$$

For SHY and IEF, a symmetric scaling is sufficient, so  $\beta_+ = \beta_-$ . For LQD, HYG, TIP, and CWB, positive and negative scaling can differ, which helps match drawdown behavior without excessively distorting average returns.

The calibration parameters  $\alpha$ ,  $\beta_+$ , and  $\beta_-$  are estimated by minimizing deviations between the calibrated proxy and the observed ETF across three aggregate statistics:

$$\text{CAGR}, \quad \sigma_{\text{annual}}, \quad \text{MaxDD}. \quad (37)$$

The objective function also contains a regularization term that keeps the parameters close to  $\alpha = 0$ ,  $\beta_+ = 1$ , and  $\beta_- = 1$ . Once estimated on the ETF overlap period, the same calibration is applied to the full historical proxy.

### 3.4.9 T10Y3M regime variable

The yield-curve variable is not an investable asset. Whenever the official FRED T10Y3M series is available, it is used directly. To extend the signal over the full 1971 universe, it can also be computed from the underlying 10-year Treasury yield and 3-month Treasury bill rate:

$$T10Y3M_t = GS10_t - TB3MS_t. \quad (38)$$

Thus, the final monthly bond data set contains six investable series, SHY, IEF, LQD, HYG, TIP, and CWB, plus the non-investable T10Y3M regime variable. The data sources for the underlying reconstruction include Treasury yields, corporate yields, CPI, real yields, and the yield-curve spread from FRED [6, 7, 8, 9, 10, 11, 12, 5].

### 3.5 Common Preprocessing Rules

All series are checked for missing observations and aligned on month-end dates. For individual stocks, abnormal adjusted-price histories are filtered by detecting very large close-to-close ratios. When such a discontinuity is detected, the price history before the discontinuity is excluded from the active sample. This avoids using histories affected by bad adjusted data, symbol changes, or non-comparable price series.

## 4 External Strategy Specification

The detailed rules of the four allocation blocks are not reproduced in this document. The Momentum Stocks, Gold, Sector, and Bond strategies are defined in the companion document *Momentum Multi Strategy* available in the project repository [1]. That external specification is the reference for:

- the investable universes and regional stock-selection process;
- the monthly signal definitions and ranking logic;
- the asset-allocation decisions inside each block;
- the rebalance timing convention and cash-handling rules;
- the implementation choices used by the backtest code.

The present report uses the return streams produced by those rules and focuses on the data reconstruction problem: extending the underlying inputs far enough back in time to evaluate the blocks over comparable long samples. Consequently, formulas are provided for data construction, index reconstruction, splicing, calibration, and portfolio aggregation, but not for the internal rule set of each strategy block.

## 5 Composite Portfolio Construction

The combined portfolio return is calculated as the weighted sum of the four allocation-block returns:

$$r_{\text{combo}}(t) = 0.50 r_{\text{MOM}}(t) + 0.10 r_{\text{Gold}}(t) + 0.10 r_{\text{Sector}}(t) + 0.30 r_{\text{Bond}}(t). \quad (39)$$

The combined equity curve is obtained by compounding these monthly returns:

$$E_{\text{combo}}(t) = E_{\text{combo}}(t-1) [1 + r_{\text{combo}}(t)]. \quad (40)$$

The combination starts in January 1980, which is the first common start date for all four long-history strategy blocks.

## 6 Backtest Procedure and Results

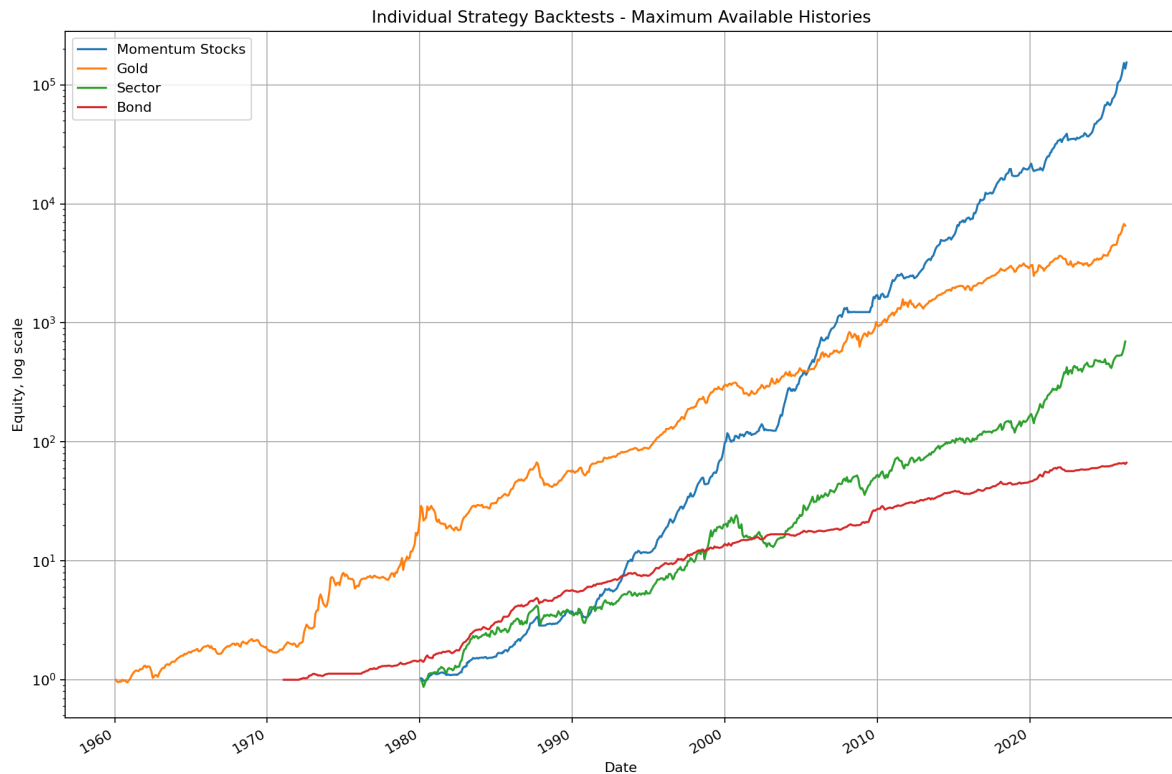
The individual strategy backtests use the longest available history for each data set:

- Momentum Stocks: consolidated regional stock momentum history from 1980, based on EODHD adjusted stock histories;

- Gold: reconstructed gold and S&P 500 total return history from 1960;
- Sector: reconstructed sector-timing history from 1980;
- Bond: reconstructed fixed-income history from 1971.

The combined portfolio backtest starts in January 1980 because all four allocation blocks are available from that date. Performance is calculated from monthly returns. The reported mean return and volatility are annualized from monthly data, and the Sharpe ratio assumes a zero risk-free rate.

## 6.1 Maximum-History Individual Backtests

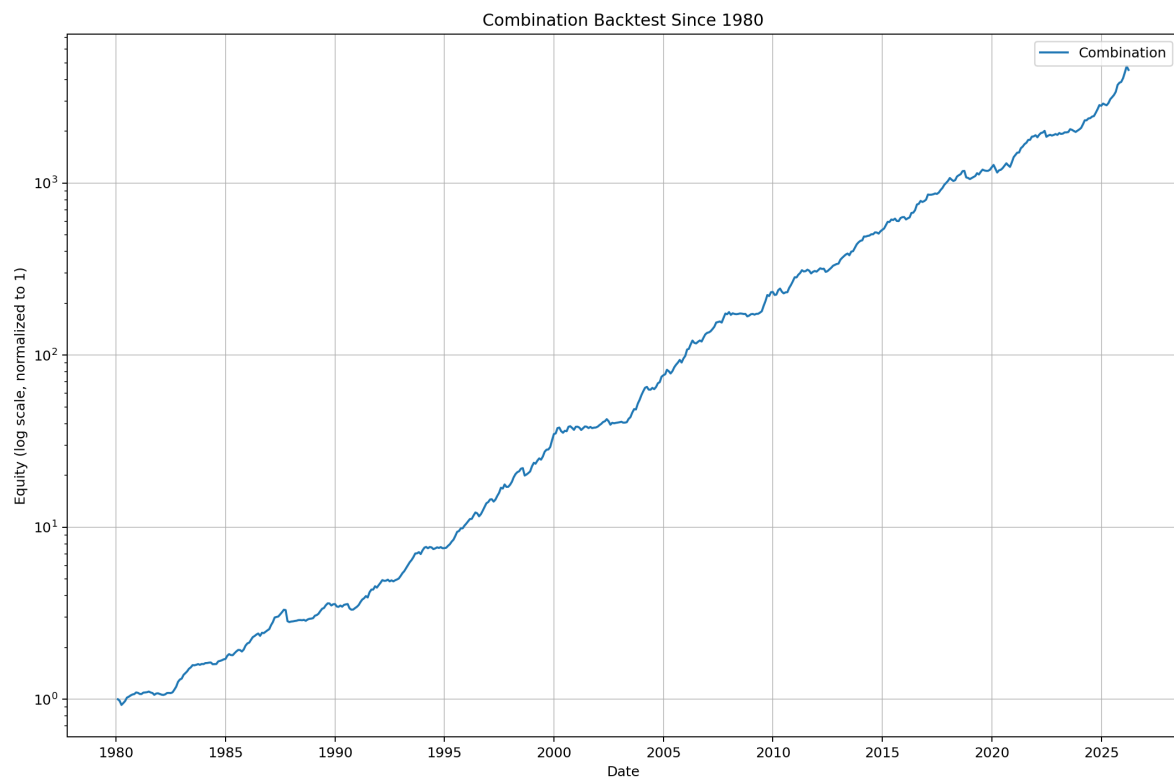


**Figure 2:** Individual strategy equity curves over their maximum available histories, log scale.

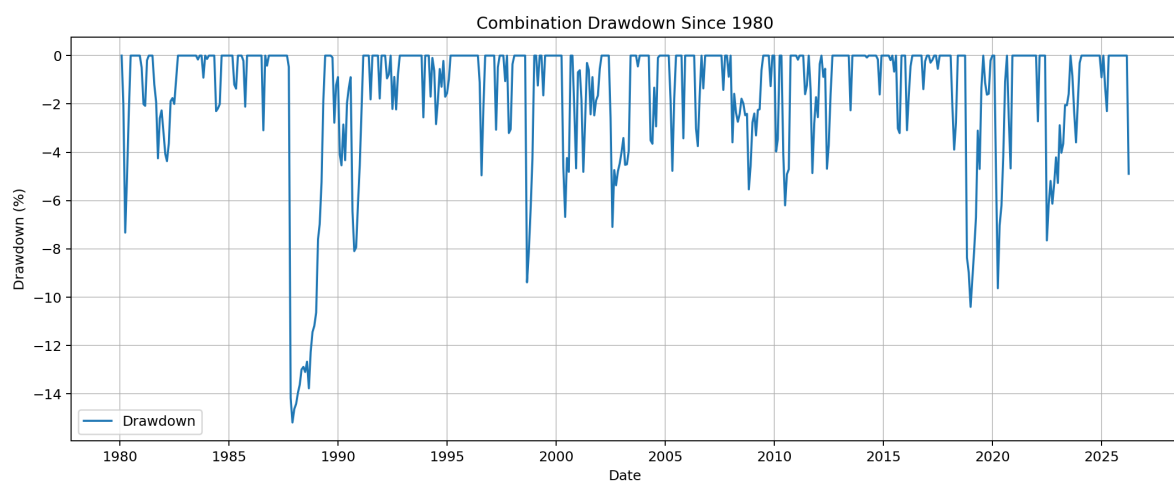
**Table 4:** Individual strategy performance over maximum available histories.

| Strategy        | Start      | End        | Years | Mean   | Stdev  | Sharpe | MaxDD  |
|-----------------|------------|------------|-------|--------|--------|--------|--------|
| Momentum Stocks | 1980-01-31 | 2026-04-30 | 46.2  | 26.03% | 14.54% | 1.79   | 16.21% |
| Gold            | 1960-01-31 | 2026-03-31 | 66.2  | 14.74% | 16.87% | 0.87   | 37.95% |
| Sector          | 1980-01-31 | 2026-03-31 | 46.2  | 15.88% | 17.95% | 0.88   | 46.06% |
| Bond            | 1971-01-31 | 2026-04-30 | 55.2  | 7.77%  | 5.47%  | 1.42   | 10.25% |

## 6.2 Combination Since 1980



**Figure 3:** Combined portfolio equity curve from January 1980, log scale.



**Figure 4:** Combined portfolio drawdown from January 1980.

**Table 5:** Combined portfolio performance from January 1980.

| Metric                  | Value      |
|-------------------------|------------|
| Start date              | 1980-01-31 |
| End date                | 2026-03-31 |
| Annualized mean return  | 18.85%     |
| Annualized volatility   | 9.68%      |
| Annualized Sharpe ratio | 1.95       |
| Maximum drawdown        | 15.18%     |
| CAGR                    | 20.01%     |
| Total growth multiple   | 4545.5x    |

The 1980–2026 combined backtest shows a high long-run compounded return with a Sharpe ratio close to 2. The drawdown profile remains controlled relative to the equity-oriented components, mainly because the final portfolio combines geographically diversified stocks, gold exposure, sector exposure, and a defensive fixed-income block.

## 7 Strategy Summary

**Table 6:** Allocation blocks and external rule reference.

| Block           | Weight | Frequency | Rule source                                          |
|-----------------|--------|-----------|------------------------------------------------------|
| Momentum Stocks | 50%    | Monthly   | Defined in the companion strategy specification [1]. |
| Gold            | 10%    | Monthly   | Defined in the companion strategy specification [1]. |
| Sector          | 10%    | Monthly   | Defined in the companion strategy specification [1]. |
| Bond            | 30%    | Monthly   | Defined in the companion strategy specification [1]. |

This separation keeps the report focused on historical-data construction and on the backtest synthesis. The strategy rules should be read from the companion PDF, while the present document should be used to audit the input histories, the synthetic extensions, the validation diagnostics, and the combined performance results.

## References

- [1] Denis Laurichesse, *Momentum Multi Strategy*, project PDF specification. [https://github.com/denis-laurichesse/momentum\\_stock\\_trading/blob/main/momentum\\_multi\\_strategy.pdf](https://github.com/denis-laurichesse/momentum_stock_trading/blob/main/momentum_multi_strategy.pdf)
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- [7] Federal Reserve Bank of St. Louis, FRED series GS10, 10-Year Treasury Constant Maturity Rate. <https://fred.stlouisfed.org/series/GS10>

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- [8] Federal Reserve Bank of St. Louis, FRED series AAA, Moody's Seasoned Aaa Corporate Bond Yield. <https://fred.stlouisfed.org/series/AAA>
  - [9] Federal Reserve Bank of St. Louis, FRED series BAA, Moody's Seasoned Baa Corporate Bond Yield. <https://fred.stlouisfed.org/series/BAA>
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  - [11] Federal Reserve Bank of St. Louis, FRED series DFII10, 10-Year Treasury Inflation-Indexed Security, Constant Maturity. <https://fred.stlouisfed.org/series/DFII10>
  - [12] Federal Reserve Bank of St. Louis, FRED series TB3MS, 3-Month Treasury Bill Secondary Market Rate. <https://fred.stlouisfed.org/series/TB3MS>